

Recommendations for the National AI Action Plan 2025

Submitted by MARTA ALI STUDIOS

*To Networking and Information Technology Research and Development (NITRD) ,
National Coordination Office (NCO), National Science Foundation.*

Greetings,

Thank you for giving us the opportunity to provide input towards shaping the **AI Action Plan** to ensure that the United States of America continues to be a global leader in AI.

As an **innovative design and fabrication company** working across the **building and manufacturing sectors**, we are focused on using **AI to enhance productivity, streamline complex processes, and accelerate innovation**. Through our initiative, “**Designing the Future Now**”, we have been also actively engaged in AI innovation, implementation and industry advocacy efforts at the **local, national, and international levels**.

Based on our experience, we believe any successful AI strategy must move beyond broad national directives. **Effective implementation starts at the local level**—in neighborhoods, cities, and states—and should scale regionally and nationally. This kind of **tiered approach ensures relevance, accessibility, and measurable outcomes**, especially for industries and communities still navigating the value, application and effective implementation of AI.

The following **strategic recommendations** reflect this view and are intended to support America’s leadership in AI while driving real impact across sectors.

“AI Action Plan” Strategic Focus Areas

1. Sectoral Transformation (Top Priority)

- **Develop AI Roadmaps by Sector:** Provide clear guidance for industries such as — **building , manufacturing, infrastructure, healthcare, finance**—on how to integrate AI, with measurable outcomes and scalable solutions. For instance ;
 - **Modernize the Building Industry:** Support AI adoption to reduce **regulatory bottlenecks**, increase **efficiency in design and construction**, **reduce risk and liability**, and enable smart and **sustainable development**. Also , encourage adoption of emerging

technologies like **generative design, predictive analytics, and digital twins** to help transform the sector forward more rapidly.

- **Advance Manufacturing with AI:** Encourage the use of AI in **advanced manufacturing** to enable faster product development, smart materials innovation, and more optimised supply chains.
- **Close the Adoption Gap:** Many industry professionals are still uncertain about AI's value or lack resources to adopt it effectively. Provide **sector-specific education, pilot programs, and hands-on support** to address this gap.
- **Lead with Intention:** Without clear national leadership, AI adoption risks being uneven. A structured, intentional approach will ensure all sectors move forward at a more progressive and uniform pace.

2. Education and Workforce

- **AI Education at Every Level:** Roll out AI education and awareness programs at the **local, city, state, regional, and national levels**. Several communities also need funding support to try out and implement AI since not all relevant AI tools are accessible without hiking subscription fees.
- **Reskill and Upskill:** Launch programs to help workers in fast-changing industries gain **AI-related skills**, using AI itself to offer **personalized learning** could be one option.
- **Focus on Critical Thinking:** AI should be a tool to **support—not replace—critical thinking**. Integrate **problem-solving and creative reasoning** into all AI training.
- **Equip Educators with the right AI tools:** Provide training and resources to help educators teach both **technical and ethical dimensions** of AI.
- **Fellowship Programs:** Create local, regional and national fellowship programs to accelerate the **development of AI talent** and help shape leadership across sectors.

3. Research & Development

- **Invest in R&D:** consider strong investment in **AI research**, especially in areas like **secure AI, generative models**, and applications for sectors like **manufacturing , building and infrastructure sectors**.
- **AI Centers of Excellence:** Create hubs that bring together **researchers, industry, and government** to develop practical, high-impact AI solutions.
- **Support Partnerships:** Incentivize **collaboration** between public and private sectors to bring AI research to market.

4. Innovation and Competitiveness

- **Support AI Startups:** Offer **grants, tax incentives, and regulatory support** to help AI startups grow, especially in **underserved areas**.
- **Modernize Traditional Industries:** Fund **pilot programs** to show how AI can transform sectors like the **building, manufacturing and infrastructure**.
- **Establish better Standards:** Take the lead on **global AI standards and ethics**, aligning with U.S. values.

In conclusion, in order for the U.S.A. to lead in AI, we need a **clear, focused strategy** that works at every level—from local communities to national policy—and across all sectors. AI adoption must be **intentional, inclusive, practical and robust**, supporting industries and empowering communities across the nation.

We are also committed to contributing to this effort and helping drive meaningful, measurable outcomes as AI continues to shape the future of our nation through initiatives like “**Designing the Future Now.**”

Thank you for your consideration!

Sincerely,

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